

→ whenever we discuss model prediction we have to understand prediction errors (bias and variance).

→ Gaining a proper understanding of these errors helps us to build accurate models but also avoid the mistakes as overfitting and underfitting.

→ Bias ?

Bias is the difference between the average prediction of our model and the correct value which we are trying to predict. Models with high bias pays very little attention to the training data and oversimplifies the model.

Variance ?

Variance is a value which tells us the spread of our data. Models with high variance pays a lot of attention to the training data and do not generalize on the data which it hasn't seen before.

Mathematically :-

Lets assume that the relationship between the predicted y and x is defined as

$$y = f(x) + e$$

where $e \rightarrow$ error

$f(x) \rightarrow$ function of x

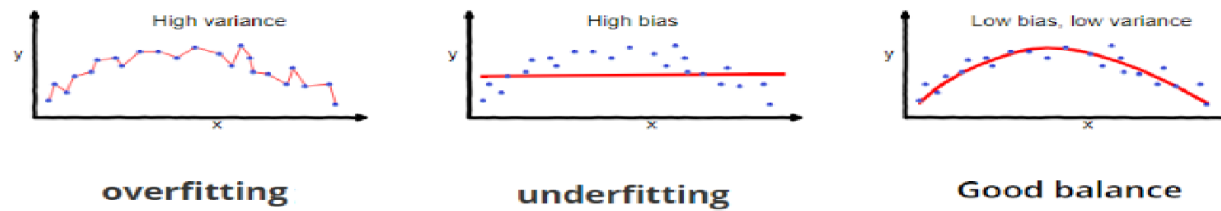
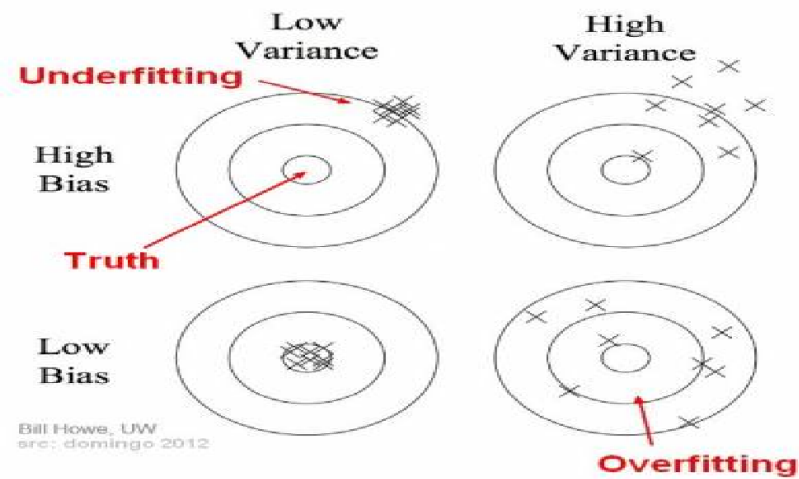
After making the model $\hat{f}(x)$ of $f(x)$ using regression or other technique we can calculate the error as.

$$Err(x) = E[(y - \hat{f}(x))^2]$$

The $Err(x)$ is further decomposed into:

$$Err(x) = (E[\hat{f}(x)] - f(x))^2 + E[(\hat{f}(x) - E[\hat{f}(x)])^2] + \sigma_e^2$$

$$Err(x) = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$



$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

